Computational Statistics and Simulation

Worksheet 3: MCMC Methods

Academic year 2025/26

1. Consider the Rayleigh distribution with probability density function

$$f(x) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)}, \qquad x \geq 0,\ \sigma > 0.$$

Use the Metropolis-Hastings algorithm to generate values from this distribution, taking as proposal function $q(y \mid x_t)$ the chi-square distribution with x_t degrees of freedom. With $\sigma = 4$ and 20000 replications, check whether the chain is efficient.

2. The aim is to generate values from a target Student's t distribution with n degrees of freedom, using the random-walk Metropolis-Hastings algorithm with the normal distribution $N(0, \sigma^2)$ for the random error.

   (a) Implement the Metropolis algorithm described, considering a chain of length $m = 10000$, $\sigma = 0.05$ and 4 degrees of freedom for the Student's t. Analyse its efficiency.

   (b) Compare the results produced by the algorithm in terms of the number of rejections while varying σ^2, taking $\sigma^2 = 0.05$; 0.5; 2; 16.

   (c) Compare the quantiles of the theoretical (target) distribution — Student's t with 4 d.f. — with the quantiles of the distribution of the generated values, for each of the values of σ^2 in the previous item and using a burn-in period of 1000. Draw conclusions.

3. Consider a random sample $(Z_1, \ldots, Z_n)$ from the following mixture of Normals:

$$p\, N(\mu_1, \sigma_1^2) + (1 - p)\, N(\mu_2, \sigma_2^2).$$

Assuming there is no information about p, the goal is to estimate the value of p using a $Be(1, 1)$ distribution as the proposal function g, and assuming that the point y generated in the Metropolis-Hastings algorithm is accepted with probability

$$\alpha = \min\left(1, \frac{f(Y)g(X_t)}{f(X_t)g(Y)}\right)$$

— this corresponds to the independence sampler.

Obtain an estimate of p assuming that the generated values come from the mixture $0.2\, N(0, 1) + 0.8\, N(5, 1)$.

4. According to a genetic model, animals of a given species are distributed over 4 categories with probabilities

$$p_1 = \frac{2 + \theta}{4}, \quad p_2 = \frac{1 - \theta}{4}, \quad p_3 = \frac{1 - \theta}{4}, \quad p_4 = \frac{\theta}{4},$$

where $0 \leq \theta \leq 1$ is an unknown parameter about which inference is to be made. Assume that

- a beta prior distribution with parameters (a, b) is adopted for θ;
- for a sample of size N, y_i animals were observed in the i-th category ($i = 1, \ldots, 4$, $\sum y_i = N$).

The aim is to simulate from this distribution using the Metropolis-Hastings algorithm. If we use, for example, $q(\theta_i, \theta_j) = 1$, this means that we simulate from uniforms and accept the simulated value with probability

$$\alpha(\theta_i, \theta_j) = \min\left\{1, \frac{\pi(\theta_j \mid y)}{\pi(\theta_i \mid y)}\right\}.$$

Apply this algorithm to simulate from the distribution

$$\pi(\theta \mid y) \propto (2+\theta)^{y_1}(1-\theta)^{y_2-y_3+b-1}\theta^{y_4+a-1}, \qquad 0 \leq \theta \leq 1,$$

taking $a = b = 1$.

5. In a bivariate normal model with mean vector (μ_1, μ_2), variances (σ_1^2, σ_2^2) and correlation ρ, consider the full conditional distributions

$$X_1 \mid X_2 \sim N\left(\mu_1 + \frac{\rho\sigma_1}{\sigma_2}(x_2 - \mu_2),\ (1-\rho^2)\sigma_1^2\right),$$

$$X_2 \mid X_1 \sim N\left(\mu_2 + \frac{\rho\sigma_2}{\sigma_1}(x_1 - \mu_1),\ (1-\rho^2)\sigma_2^2\right).$$

Use the Gibbs sampler to generate values from this bivariate normal, taking the parameter values $(\mu_1 = 0, \mu_2 = 2)$, $(\sigma_1^2 = 1, \sigma_2^2 = 0.25)$ and $\rho = -0.75$, and estimate the parameters through the corresponding sample characteristics.

6. Suppose that $X = (X_1, \ldots, X_n)$ is a sequence of random variables of the form $X_i = R_i Y_i$, where the Y_i are i.i.d. $Poi(\lambda)$, the R_i are i.i.d. $Ber(p)$, and Y_i and R_i are independent.

Given an observed sample $x = (x_1, \ldots, x_n)$ of X, the goal is to estimate the parameters (λ, p). From the Bayesian point of view, the following hierarchical model is considered:

- $p \sim U(0,1)$;
- $\lambda \mid p \sim Ga(a, b)$;
- $(r_i \mid p, \lambda) \sim Ber(p)$, independent;
- $(x_i \mid r, p, \lambda) \sim Poi(\lambda r_i)$, independent,

with $r = (r_1, \ldots, r_n)$ and a, b known constants. It follows that the joint distribution of (x, r, λ, p) is

$$f(x, r, \lambda, p) = \frac{b^a \lambda^{a-1}\exp(-b\lambda)}{\Gamma(a)} \prod_{i=1}^{n} \frac{\exp(-\lambda r_i)(\lambda r_i)^{x_i}}{x_i!}\, p^{r_i}(1-p)^{1-r_i}.$$

Simulations on p and λ can be carried out by simulating from the posterior distribution — the distribution of (r, λ, p) conditional on the observed data x — given by

$$f(r, \lambda, p \mid x) = \frac{f(x, r, \lambda, p)}{\int f(x, r, \lambda, p)\, d(r, \lambda, p)},$$

using, for example, the Gibbs algorithm.

(a) The full conditionals have the form

$$(\lambda \mid p, r, x) \sim Ga\left(a + \sum_i x_i,\ b + \sum_i r_i\right),$$

$$(p \mid \lambda, r, x) \sim Be\left(1 + \sum_i r_i,\ n + 1 - \sum_i r_i\right),$$

$$(r_i \mid \lambda, p, x) \sim Ber\left(\frac{p\, e^{-\lambda}}{p\, e^{-\lambda} + (1-p)\mathbb{I}_{\{x_i=0\}}}\right).$$

Prove the first two expressions.

(b) Generate a sample of size $n = 100$ from the ZIP model using the parameters $p = 0.3$ and $\lambda = 2$.

(c) Implement the Gibbs algorithm to simulate from $f(r, \lambda, p \mid x)$ for the data of the previous item, obtain estimates of the parameters $p = 0.3$ and $\lambda = 2$, and characterise the distributions of $p \mid x$ and $\lambda \mid x$.

7. McCormick and Mathew (1983) examined estimation questions related to the model

$$X_t = \gamma + \rho X_{t-1} + Y_t, \qquad t \geq 1,$$

where $\gamma \geq 0$, $0 \leq \rho < 1$ are unknown parameters and the Y_t are i.i.d. random variables. The Bayesian analysis of this model when the Y_t are exponential with mean α^{-1} was considered by Pereira and Amaral Turkman (1983). For a non-informative prior distribution, the posterior distribution based on a sample of size n is

$$h(\theta \mid x) \propto \alpha^{n-1} e^{-\alpha[S_0 - \rho S_1 - (n-1)\gamma]} \, \mathbb{I}_{\Theta_n}(\theta)$$

where $\theta = (\alpha, \gamma, \rho)$, $\Theta_n = \{\theta : \alpha \geq 0,\ 0 \leq \rho < 1,\ \gamma \geq 0,\ x_t - \rho x_{t-1} - \gamma \geq 0,\ \forall t = 1, \ldots, n\}$, $S_0 = \sum_{t=2}^{n} x_t$ and $S_1 = \sum_{t=2}^{n} x_{t-1}$. Show that:

(a) The full conditional distributions are given by

i. $h(\alpha \mid x, \gamma, \rho) \sim Ga(n,\ S_0 - \rho S_1 - (n-1)\gamma)$;

ii. $h(\gamma \mid x, \alpha, \rho) \sim Exp_{left}((n-1)\alpha, \gamma^*)$, where $\gamma^* = \min(x_t - \rho x_{t-1})$;

iii. $h(\rho \mid x, \alpha, \gamma) \sim Exp_{left}(\alpha S_1, \rho^*)$, where $\rho^* = \min\left(1, \frac{x_t - \gamma}{x_{t-1}}\right)$.

Note: $f(x) \sim Exp_{left}(\beta, \delta)$ means $f(x) = \dfrac{\beta e^{-\beta(\delta - x)}}{1 - e^{-\beta\delta}}$, for $0 < x < \delta$ (left-truncated exponential).

(b) The following data are dissolved-oxygen measurements taken at the Angeja bridge from June to November 1991:

4.0	4.1	3.9	4.4	3.2	4.0	3.7	4.2	4.5	4.3	3.6	1.9
3.3	1.9	2.9	2.7	2.4	2.9	3.8	3.5	2.7	3.9	2.8	3.3
2.9	3.8	4.4	5.1	5.2	7.2	6.2	4.8	4.0	2.7	4.4	3.4
4.2	4.8	5.3	4.5	4.1	4.0	2.9	0.8	5.2	7.3	5.1	5.3
7.1	8.1	7.8	6.9	7.5	6.0	5.0	5.3	4.8	4.3	5.8	4.6
4.5	4.1	4.6	6.4	6.3	6.2	6.2	6.8	7.5	7.4	7.0	6.7
7.5	6.1	5.7	5.4	5.3	4.0	3.7	2.5	0.8	1.3	3.3	4.1
5.7	4.3	3.5	3.8	2.0	3.8	4.1	1.8	3.0	4.7	6.2	6.0
5.3	4.4	3.4	4.7	4.5	3.7	4.3	1.6	2.9	3.6	3.7	3.9
4.6	5.0	5.3	4.7	6.5	5.7	5.8	8.0	7.4	6.1	7.6	

Using this sample, implement the Gibbs algorithm, using the `CODA` package to assess convergence, in order to estimate the parameters and approximately characterise the distributions of $\alpha \mid x$, $\gamma \mid x$ and $\rho \mid x$.

8. Consider the following regression model (one-way random effects model with k factors)

$$Y_{i,j} = \mu + \alpha_i + \epsilon_{i,j}, \qquad \alpha_i \sim N(0, \sigma_\alpha^2), \quad \epsilon_{i,j} \sim N(0, \sigma^2),$$

$j = 1, \ldots, n_i$; $i = 1, \ldots, k$. The model can also be written in hierarchical form, i.e.,

$$Y_{i,j} \mid \alpha_i \sim N(\mu + \alpha_i, \sigma^2) \quad \text{with} \quad \alpha_i \sim N(0, \sigma_\alpha^2).$$

The joint density of $y \mid \mu, \alpha_1, \ldots, \alpha_k, \sigma_\alpha^2, \sigma^2$, with $y = (y_1, \ldots, y_n)$, is given by

$$\left(\frac{1}{\sigma_\alpha^2}\right)^{k/2} \exp\left[\frac{1}{2\sigma_\alpha^2}\sum_{i=1}^{k}\alpha_i^2\right]\left(\frac{1}{\sigma^2}\right)^{N_k/2} \exp\left[\frac{1}{2\sigma^2}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\mu-\alpha_i)^2\right], \quad \text{with } N = \sum_{i=1}^{k} n_i,$$

and the full conditional distributions are

$$\alpha_i \mid y, \mu, \sigma_\alpha^2, \sigma^2 \sim N\left(\frac{n_i\sigma_\alpha^2}{n_i\sigma_\alpha^2+\sigma^2}(\bar{y}_i-\mu),\ \frac{\sigma_\alpha^2\sigma^2}{n_i\sigma_\alpha^2+\sigma^2}\right), \qquad i = 1, \ldots, k,$$

$$\mu \mid y, \alpha_1, \ldots, \alpha_k, \sigma_\alpha^2, \sigma^2 \sim N\left(\bar{y}-\bar{\alpha},\ \frac{\sigma^2}{N_k}\right),$$

$$\sigma_\alpha^2 \mid y, \alpha_1, \ldots, \alpha_k, \mu, \sigma^2 \sim IG\left(\frac{k}{2}-1,\ \frac{1}{2}\sum_{i=1}^{k}\alpha_i^2\right),$$

$$\sigma^2 \mid y, \alpha_1, \ldots, \alpha_k, \mu, \sigma_\alpha^2 \sim IG\left(\frac{N_k}{2}-1,\ \frac{1}{2}\sum_{i,j}(y_{ij}-\mu-\alpha_i)^2\right),$$

where IG denotes the inverse gamma distribution.

Consider the following data, used to assess the precision of the estimated chemical composition of turnip leaves, where the leaves represent a random effect:

factors	observations			
1	3.28	3.09	3.03	3.03
2	3.52	3.48	3.38	3.38
3	2.88	2.80	2.81	2.76
4	3.34	3.38	3.24	3.26

Implement the Gibbs algorithm to obtain estimates of the parameters α_i, $i = 1, \ldots, 4$, μ, σ_α^2 and σ^2 (after assessing convergence) and some of their characteristics, and plot the corresponding full conditional distributions.